MSCPHUB MSCP Disk Server for OpenVMS Nodes

User’s Guide

MSCPHUB-UG-0116

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Preface

Intended Audience

This manual is for the system manager who installs MSCPHUB on a Linux computer and gives its disks to OpenVMS nodes. It assumes that the reader can work as root on Linux and as SYSTEM on OpenVMS, and that the OpenVMS nodes already form a VMScluster over the LAN. Nothing else is assumed.

The programmer who is to alter the server wants the *MSCPHUB Programmer's Guide* instead. This manual says nothing about how the server is built inside.

Why MSCPHUB Exists

For forty years the VMScluster has shown how computers share their disks: every node sees every disk, a disk is served by whoever has it, and the cluster goes on when a member goes down. The machines of Digital Equipment Corporation are leaving the machine rooms. The systems they ran, and the engineering culture they were made with, should not leave with them.

MSCPHUB is one part of the work to keep that heritage alive and at work. It lets OpenVMS nodes - on Alpha, on Integrity, on x86, or emulated - keep their disks on the storage of today: a file, a partition, an LVM volume of a Linux computer. OpenVMS is not changed in any way; to OpenVMS the server is a disk controller of the cluster, like an HSC.

MSCPHUB is also made the DEC way, on purpose. Its messages have the OpenVMS form and severities, its parameters are qualifiers, it has a HELP library, its versions are IDENTs, its sources follow the conventions of the OpenVMS sources, and this manual is laid out like the manuals of OpenVMS. This is not decoration. A documented decision for everything, a message for every condition, nothing that the operator has to guess - that way of working is part of the heritage, and it is kept by being used.

Document Structure

This manual contains the following chapters and appendixes:

Part	Contents
Chapter 1	Introduces the server: what it does, how an OpenVMS node sees it, what it does not do
Chapter 2	Installing the server
Chapter 3	Configuring the server: the facts of the cluster, the disks, the parameter file
Chapter 4	Running the server and using its disks on OpenVMS
Chapter 5	Contains the command dictionary: the `mscphub` command and all its qualifiers
Chapter 6	Describes the format of a message and lists every message with what to do about it
Chapter 7	Troubleshooting: a symptom, what to look at, and what to do
Appendix A	Lists the files installed and used
Appendix B	Lists the limits of the server

The reader who installs the server for the first time wants Chapters 2, 3 and 4, in that order. The reader who holds a message wants Chapter 6, and the one who holds a symptom, Chapter 7.

Related Documents

- *OpenVMS Cluster Systems*
- *Guidelines for OpenVMS Cluster Configurations*
- *OpenVMS System Manager's Manual*
- *MSCPHUB Programmer's Guide*
- The manual page `mscphub(8)` and the help library of the server (`mscphub /HELP`)

Conventions

The following conventions are used in this manual:

Convention	Meaning
`#`	The prompt of the Linux shell of root. Text after the prompt is entered by the user.
`$`	The DCL prompt of an OpenVMS node. Text after the prompt is entered by the user.
italics	A term being defined, or a value to be replaced by one of your own.
`UPPERCASE`	A qualifier, a keyword, an OpenVMS command, or the identification of a message.
`[ ]`	An optional part. Do not type the brackets.
`|`	A choice of one of the items it separates.
`...`	The part before it may be repeated.
`Ctrl/C`	Hold down the Ctrl key and press C.

Qualifiers are shown in uppercase; `mscphub` takes them in any case.

Note

Sample output in this manual is reproduced as the server produces it. The first part of a line of the log - the date, the process and the routine - is left out where it does not matter.

Chapter 1. Introduction

1.1 What MSCPHUB Does

MSCPHUB is an MSCP disk server that runs on Linux. It makes files and block devices of a Linux computer available to the OpenVMS nodes on the same LAN as disks of their own. An OpenVMS node sees such a disk with `SHOW DEVICE`, initializes it with `INITIALIZE` and mounts it with `MOUNT`; `COPY`, `BACKUP` and `ANALYZE/DISK_STRUCTURE` work on it as on a local disk. Several nodes can mount one disk at the same time with `MOUNT/CLUSTER`.

Nothing is installed on OpenVMS, and no OpenVMS parameter is changed.

1.2 How an OpenVMS Node Sees the Server

The server speaks NISCA, the cluster protocol that PEDRIVER speaks on the LAN (Ethernet type 60-07). To an OpenVMS node it is a node of the cluster that has no votes and runs no connection manager - in the way an HSC controller is on a CI. It is therefore never a member of the VMScluster: it cannot take part in a quorum, it cannot hang the cluster, and a node can lose it without a state transition.

The server has a name (SCSNODE) and a system ID (SCSSYSTEMID) of its own. The OpenVMS node shows it in `SHOW CLUSTER` and in `MCR SCACP SHOW CHANNEL`, opens a virtual circuit to it, and connects its disk class driver to the disk server of MSCPHUB, `MSCP$DISK`. The disks appear on the node as `$`*n*`$DK`*cn* (for example `$42$DKA100`), *n* being the allocation class of the server.

1.3 What Is Served

Item	What the server does
Disks	Up to 8 per server (`/UNIT0` to `/UNIT7`). A disk is a Linux file or a block device: a whole disk, a partition, an LVM volume, a loop device, an iSCSI LUN.
Sectors	Devices of 512-octet and of 4096-octet sectors. OpenVMS always works in blocks of 512 octets; the server does the rest.
Read-only disks	A disk can be served read only - for example the CD image of OpenVMS, as an RRD42.
Commands	READ, WRITE, ERASE, COMPARE HOST DATA, ONLINE, AVAILABLE, GET UNIT STATUS and the other MSCP commands an OpenVMS disk class driver sends.
Nodes	Every OpenVMS node of the group on the LAN, at the same time.

1.4 When a WRITE Is Done

OpenVMS never asks a disk to empty a cache of its own: when a disk says that a WRITE is done, OpenVMS takes it that the data is on the medium. The server therefore answers a WRITE only when its data is on the disk of the Linux computer. This is the default and it is safe against a crash or a power failure of the Linux computer. The qualifier `/UNSAFE` turns it off, for tests only.

1.5 What MSCPHUB Does Not Do

- It is not a member of the VMScluster and has no votes.
- It serves disks only: no tapes, and no boot of diskless satellites.
- It cannot yet be a member disk of a shadow set.
- It works over the LAN only. A cluster over IP (IPCI) is not supported.
- It uses one LAN adapter.

MSCPHUB is verified with OpenVMS Alpha Version 8.3. Other versions that cluster over the LAN with PEDRIVER are expected to work.

Chapter 2. Installation

2.1 Requirements

- A Linux computer with a kernel of the last five years (5.10 or later; Debian 11 or later).
- An Ethernet connection to the LAN of the OpenVMS nodes: the same switch, VLAN or bridge.
- The StarLet package, version 1.6 or later, installed.
- The packages `cmake`, `gcc`, `pkg-config` and `liburing-dev`.
- Optionally the HELP package: with it, `/HELP` displays the whole help library.
- root on Linux.

2.2 Building and Installing

MSCPHUB is built from its kit, `MSCPHUB-`*ident*`.tar.gz`:

```
# apt install cmake gcc pkg-config liburing-dev
# tar xzf MSCPHUB-X01-16.tar.gz
# cd MSCPHUB-X01-16
# cmake -S . -B build
# cmake --build build -j
# ctest --test-dir build
# cmake --install build
```

Install only when `ctest` says `100% tests passed`.

2.3 What Is Installed

With the default prefix, `/opt/mscphub`:

File	What it is
`/opt/mscphub/sbin/mscphub`	The server
`/etc/opt/mscphub/mscphub.conf.example`	A sample parameter file
`/lib/systemd/system/mscphub.service`	The systemd unit; not enabled
`/opt/mscphub/share/man/man8/mscphub.8`	The manual page
`/opt/mscphub/share/help/mscphub.hlb`	The help library (with the HELP package)
`/opt/mscphub/share/doc/mscphub/`	The manuals and README

2.4 Search Paths

The server and its manual page are not on the search paths of the shell. Add them, for example in `/root/.profile`:

```
export PATH=$PATH:/opt/mscphub/sbin
export MANPATH=$MANPATH:/opt/mscphub/share/man
```

2.5 Installing in Another Place

Give the prefix when the build is configured:

```
# cmake -S . -B build -DCMAKE_INSTALL_PREFIX=/usr/local
```

The parameter file is then in `/usr/local/etc`.

2.6 Verifying the Installation

```
# mscphub /HELP
```

displays the description of the server. No OpenVMS node is needed for it.

2.7 Removal

From the build directory of the installation:

```
# cmake --build build --target uninstall
```

This removes what that installation put in place, and nothing else. The parameter file `mscphub.conf` is yours and is left.

Chapter 3. Configuration

3.1 The Facts of the Cluster

Write down four things on one OpenVMS node of the cluster.

The group number:

```
$ MCR SYSMAN CONFIGURATION SHOW CLUSTER_AUTHORIZATION
```

The password hash. It is in `SYS$SYSTEM:CLUSTER_AUTHORIZE.DAT`:

```
$ DUMP SYS$SYSTEM:CLUSTER_AUTHORIZE.DAT
```

Take the quadword at offset 8 and write its octets from the lowest one. The longwords `00000000 8EF54F24` there give `/AUTHORIZE=5A3C9E1700000000`.

The names and the system IDs in use:

```
$ SHOW CLUSTER
$ MCR SYSGEN SHOW SCSSYSTEMID
```

Choose for the server a name (`/NODE`, up to 6 characters) and a system ID (`/SYSTEMID`, 1 to 65535) that no node of the cluster has. The server checks this itself when it starts, and stops if it finds a clash.

An allocation class for the disks, 1 to 255, that the cluster does not use for disks of the same names. With `/ALLOCLASS=42` the disk `DKA100` is `$42$DKA100:` on every OpenVMS node.

3.2 Preparing the Disks

3.2.1 A File

A new empty file of 2 GB:

```
# mkdir -p /srv/mscphub
# truncate -s 2G /srv/mscphub/dka100.img
```

Such a file is *sparse*: it takes no space yet, and the first write to every part of it is slower. For a disk that must be fast, write the file full once instead:

```
# dd if=/dev/zero of=/srv/mscphub/dka100.img bs=1M count=2048
```

The size of the file when the server starts, in blocks of 512 octets, is the size of the disk.

Put the files on ext4 or XFS. The server reads and writes them with direct I/O, as it does a block device. On a file system without direct I/O (tmpfs of an old kernel, some network file systems) it uses the page cache and says `NODIRECT`; the disk works, but every write costs more.

3.2.2 A Block Device

A whole disk, a partition, an LVM volume, a loop device or an iSCSI LUN. It must not be mounted or used by Linux: the server takes it for itself and refuses one that is in use (`DEVBUSY`). A partition or an LVM volume is a little faster than a file.

3.2.3 A Disk of 4096-Octet Sectors

Such a disk works. A partition on it must start on a physical sector, or every write is slow (the server says `MISALIGN`); `parted` and `fdisk` start partitions on 1 MB by default. Section 4.3 says how to initialize such a disk on OpenVMS.

3.2.4 A CD Image

The CD of OpenVMS, an ISO file, is served read only, as an RRD42:

```
/UNIT2=DKA300,/srv/mscphub/AXPVMS083.iso,RO
```

Caution

> Never give one file or device to two servers, and never mount it on Linux while it is served.

3.3 The Parameter File

```
# cp /etc/opt/mscphub/mscphub.conf.example /etc/opt/mscphub/mscphub.conf
# vi /etc/opt/mscphub/mscphub.conf
```

The file holds one qualifier on a line, as on the command line. A line that begins with `!` is a comment. A complete example:

```
! MSCPHUB of our cluster
/INTERFACE=eth1
/NODE=MHUB
/SYSTEMID=65200
/GROUP=4242
/AUTHORIZE=5A3C9E1700000000
/ALLOCLASS=42
/UNIT0=DKA100,/srv/mscphub/dka100.img
/UNIT1=DKA200,/dev/vg0/vmsdata
/UNIT2=DKA300,/srv/mscphub/AXPVMS083.iso,RO
```

Every qualifier is described in Chapter 5.

Note

A qualifier the server does not know is passed over with a warning (`Skip unrecognized option`), and the server runs without it. Read the log after every change of the file.

3.4 Checking the Configuration

Start the server by hand with `/PASSIVE`. It only listens and sends nothing, so it is safe on any LAN:

```
# mscphub /CONFIG=/etc/opt/mscphub/mscphub.conf /PASSIVE
```

After the 4 seconds of `/LISTEN` there is a `NEWNODE` for every OpenVMS node. Press `Ctrl/C`. No `NEWNODE`: see Chapter 7.

Chapter 4. Operation

4.1 Starting the Server

```
# systemctl enable --now mscphub
# journalctl -u mscphub -f
```

The unit is not started at boot until you enable it, and it does not start without `/etc/opt/mscphub/mscphub.conf`. This is on purpose: a server that starts by itself talks to the cluster on its LAN at once.

4.2 What the Log Shows at Start

In this order:

```
%MSCPHUB-I-UNITCFG, unit $42$DKA100 is /srv/mscphub/dka100.img, 4194304 blocks, a file, sector 512, RZ29, S
%MSCPHUB-I-STARTUP, MSCPHUB X01-16, node MHUB, system ID 65200, group 4242, on eth1 (8E-
4E-65-DB-5C-B3)
%MSCPHUB-I-LISTEN, listening 4 seconds before announcing the node
%MSCPHUB-I-NEWNODE, node ALPHA1, system ID 65534, adapter 08-00-2B-E5-40-01, CC V1.6 (104 octets)
%MSCPHUB-I-ANNOUNCE, node MHUB announced on eth1, 1 other node seen in group 4242
%MSCPHUB-I-CHOPEN, channel to node ALPHA1 (65534) is open, CC V1.6, pipe quota 16, buffer 1426, services 000
%MSCPHUB-I-VCOPEN, virtual circuit to node ALPHA1 (65534) is open: VMS V8.3 on ALPH, datagram 576, message 2
%MSCPHUB-I-CONOPEN, connection VMS$DISK_CL_DRVR of ALPHA1 (65534) to MSCP$DISK is open, credit 10/16
```

`CONOPEN` means that the disk class driver of the node has connected: the disks are there.

4.3 Using the Disks on OpenVMS

A few seconds after `CONOPEN`:

```
$ SHOW DEVICE $42$DKA
$ INITIALIZE $42$DKA100: DATA1
$ MOUNT/SYSTEM $42$DKA100: DATA1
$ COPY *.* $42$DKA100:[000000]
```

On a Linux disk of 4096-octet sectors, give the cluster size yourself, 8 or a multiple of 8:

```
$ INITIALIZE/CLUSTER=16 $42$DKA100: DATA1
```

Then the files fill whole sectors and are written without being read first. OpenVMS takes 16 by itself on a disk of more than 50000 blocks, and 1 on a smaller one.

Two or more nodes can use one disk together:

```
$ MOUNT/CLUSTER $42$DKA100: DATA1
```

4.4 Displaying the State of the Server

```
# systemctl kill -s USR1 mscphub
# journalctl -u mscphub -n 40
```

On `SIGUSR1` the server writes to the log, in this order:

Message	What it reports
`UPTIME`	The version, the name, how long the server has run
`NODESTATS`	One for each node: the channel, the virtual circuit, the connections open
`STATS`	The frames received and sent
`TRSTATS`	The transport: packets, retransmissions, duplicates
`ACKSTATS`	How the packets of the nodes were acknowledged
`BLKSTATS`	The data moved
`MSCPSTATS`	Commands in progress and the data buffers in use
`UNITSTATS`	One for each disk: hosts, commands, reads and writes
`BKSTATS`	One for each disk: the work of the Linux disk
`LATSTATS`	One for each disk and for each of READ and WRITE: how long the commands took

The counters run from the start of the server. To measure a test, send `SIGUSR1` before and after it and subtract. Chapter 6 describes each message.

4.5 Stopping the Server

Dismount its disks on OpenVMS first. Then:

```
# systemctl stop mscphub
```

The server says `HOSTSHUT` to every node and `BYE` to the group; the OpenVMS nodes close their circuits to it at once. Last, it writes its state and its counters to the log, as on `SIGUSR1`.

A disk still mounted when the server stops waits for it - the node tries again - and the work goes on when the server comes back.

4.6 Altering the Service

To pin the server to a CPU, or to change its start line, use a drop-in, not the unit file:

```
# systemctl edit mscphub
[Service]
CPUAffinity=3
```

4.7 Backing Up the Disks

Back up an image file as you back up any disk: with the server stopped, or with the disk dismounted on all OpenVMS nodes. A copy of a file that a node is writing is not a consistent disk.

Chapter 5. Command Dictionary

This chapter describes the `mscphub` command and each of its qualifiers.

A qualifier is given on the command line or, one on a line, in the file of `/CONFIG`. `--NAME=value` may be written for `/NAME=value`. A qualifier the server does not know is passed over with a warning.

MSCPHUB

Starts the MSCP disk server.

Format

```
mscphub /CONFIG=file-spec
mscphub /INTERFACE=name /NODE=name /SYSTEMID=n /GROUP=n /AUTHORIZE=hex [/qualifier...]
mscphub /HELP["topic..."]
```

Parameters

None.

Description

The server announces itself on the LAN of `/INTERFACE` after `/LISTEN` seconds, and checks that no node already uses its name or its system ID. It opens a channel and a virtual circuit to every OpenVMS node of the group, and serves the disks of `/UNIT0` to `/UNIT7` to every node that connects to its disk server. It runs until it receives `SIGINT` or `SIGTERM`.

The server needs `CAP_NET_RAW`: run it as root. The qualifiers `/INTERFACE`, `/NODE`, `/SYSTEMID`, `/GROUP` and `/AUTHORIZE` are required.

The signals the server takes:

Signal	Action
`SIGINT`, `SIGTERM`	Stop the server (message `SHUTDOWN`)
`SIGUSR1`	Write the state and the counters to the log (Section 4.4)

The image returns 0 to the shell when it stops on a signal, and 1 when it stops on a condition of severity F.

Qualifiers

/ACK={`HOLD` | `FLUSH`} Specifies when the server acknowledges the packets of a node in a packet of its own. Most of the time the acknowledgement travels in a packet of the server - the answer to a command, the data of a READ. When the server has nothing to send, it sends the acknowledgement alone.

HOLD sends the acknowledgement alone only when the node asks for it at once, after 4 packets, or after 20 ms at most; otherwise it waits for the answer of the server. FLUSH sends it as soon as the server has taken all the frames that came in.

The default is /ACK=HOLD. HOLD sends far fewer packets; the messages ACKSTATS and LATSTATS compare the two.

/ALLOCLASS=*n* Specifies the allocation class of the server, 0 to 255. With /ALLOCLASS=42 the OpenVMS nodes name the disk DKA100 of the server `$42$DKA100`; with 0 they name it *node*`$DKA100`, after the SCSNODE of the server. The default is 0.

Do not change it while OpenVMS nodes use the disks, and do not give a class that the cluster uses for disks of the same names.

/AUTHORIZE=*hex* Specifies the cluster password, as the 8-octet hash OpenVMS keeps in `SYS$SYSTEM:CLUSTER_AUTHORIZE.DAT`: 16 hexadecimal digits, the octets from the lowest one. Required. Section 3.1 says how to read it.

/CONFIG=*file-spec* Reads the qualifiers from the file named, one on a line. A line that begins with ! is a comment. The qualifiers of the command line are taken too.

/CREDITS=*n* Specifies how many commands each OpenVMS node may have in progress on the disk server at a time, 2 to 64. The value is given to a node when it connects. The default is 16.

/GROUP=*n* Specifies the cluster group number, 1 to 65279. Required.

/HELP[=*"topic..."*] Displays the help library of the server: the whole description, or the topic named, and stops.

/INTERFACE=*name* Specifies the Linux interface the OpenVMS nodes are on: an Ethernet adapter, a bridge, a VLAN, a tap device in a bridge. Required. The MAC address of the interface is not changed.

/LISTEN=*seconds* Specifies how long the server listens before it sends anything, 1 to 60 seconds. In that time it learns the nodes of the LAN from their HELLO messages; a node that uses the name or the system ID of the server stops it. The default is 4.

/LOGFILE=*file-spec* Writes the messages to the file named. By default they go to the standard error, which systemd keeps in its journal.

/NODE=*name* Specifies the SCSNODE of the server, 1 to 6 of the characters A-Z, 0-9, $ and _. Required. It must not be the name of any other node of the group.

/PASSIVE The server only listens and reports the nodes it sees. It opens no channel and sends nothing at all. Safe on any LAN, for example to look at a production cluster.

/SYNC={DSYNC | GROUP | NONE} Specifies how a WRITE is made durable before it is answered.

DSYNC writes every WRITE to the medium by itself. GROUP makes the writes that end while the disk is being flushed wait for the next flush, and answers them together: fewer flushes when many nodes write at once, and as safe as DSYNC. NONE is the same as /UNSAFE.

The default is /SYNC=DSYNC.

/SYSTEMID=*n* Specifies the SCSSYSTEMID of the server, 1 to 65535. Required. It must differ from that of every other node of the group; on the LAN the nodes are told apart by the low 16 bits of the SCSSYSTEMID only.

/TIMEZONE={LOCAL | UTC | [+|-]*hh*[:*mm*]} Specifies the zone of the time the server tells the nodes when it opens a circuit. SHOW CLUSTER and SCACP show it as the time the server came up. LOCAL is the zone of the Linux computer, summer time and all. The default is LOCAL. Only the operator sees this time; the protocol works with any zone.

/TRACE Writes every frame received and every step of every channel to the log. Use it for a short time only.

/UNIT0=*name*,*path*[,RO][,MEDIA=*medium*][,SECTOR=*octets*] ... /UNIT7=... Specifies a disk. *name* is DK*cn*: a controller letter A to Z and a number 0 to 9999, for example DKA100. *path* is a Linux file or a block device (Section 3.2).

The options, in any order:

RO - read only. The node mounts the disk write-locked.

MEDIA=*medium* - what the node is told the disk is: two or three letters and a number. The default is RZ29, and RRD42 with RO.

SECTOR=*octets* - takes a file as a disk of 512, 1024, 2048 or 4096-octet sectors. For tests of 4K disks only; a block device tells its sector itself.

The server takes a file for itself with a lock, and opens a block device for exclusive use: a second server cannot serve it, and a device mounted on Linux is refused (DEVBUSY).

/UNSAFE Answers a WRITE as soon as Linux has taken the data, before it is on the disk. Faster, but a crash or a power failure of the Linux computer can lose data an OpenVMS node believes is written. For tests only. The server says `UNSAFE` when it starts.

Examples

```
# mscphub /CONFIG=/etc/opt/mscphub/mscphub.conf /PASSIVE
```

Starts the server with its parameter file, to listen only. The server reports every node it sees and sends nothing.

```
# mscphub /INTERFACE=eth1 /NODE=MHUB /SYSTEMID=65200 /GROUP=4242 /AUTHORIZE=5A3C9E1700000000 -

  /ALLOCLASS=42 /UNIT0=DKA100,/srv/mscphub/dka100.img
```

Starts the server by hand with one disk; the nodes see it as `$42$DKA100`. (The line is broken here for print; type it as one.)

```
# mscphub "/HELP=Messages IDCLASH"
```

Displays the description of the message IDCLASH.

Chapter 6. Messages

6.1 Message Format

The server reports every condition in the OpenVMS form:

```
21-10-2026  14:04:20.462  3127338  [MSCPHUB\main:503]  %MSCPHUB-I-STARTUP,  MSCPHUB  X01-16, node MHUB, ...
```

The fields are the date and time, the process ID, the module and routine that found the condition, then the facility name `MSCPHUB`, the severity, the identification and the text of the message.

The severity letters are:

Letter	Severity	Meaning
`S`	Success	The operation completed
`I`	Information	Something worth knowing; nothing to do
`W`	Warning	Something is not as it should be; the server goes on
`E`	Error	An operation failed; the server goes on
`F`	Fatal	The server stops

6.2 Message Reference

The messages are listed in alphabetical order. A word in *italics* stands for a value the server puts in; "(s)" stands for a plural ending.

ACKSTATS, information acknowledgements (/ACK=*mode*): *n* asked for at once; *n* with a packet of ours, *n* alone: *n* asked, *n* for a duplicate, *n* for *n* packets, *n* when drained, *n* on the tick

Explanation: Written on SIGUSR1 after TRSTATS. How many packets of the nodes asked to be acknowledged at once, how many acknowledgements went in a packet of the server, and how many went alone - because the node asked, for a packet that came twice, for a batch of packets (/ACK=HOLD), when all frames were taken (/ACK=FLUSH), or on the 20 ms tick. Fewer alone is better.

User Action: None.

ANNOUNCE, information node *name state* on *interface*, *n* other node(s) seen in group *n*

Explanation: The listen time is over and the server now sends HELLO messages. With /PASSIVE it says "listens only" and sends nothing.

User Action: None. If no other node is seen, see Section 7.2.

AUTHFAIL, warning node *name* (*systemid*, *adapter*) sent a CC message with a wrong cluster password, ignored

Explanation: The node is in the group of the server, but its password hash is not that of /AUTHORIZE. No channel is opened to the node. The message is written once for each node.

User Action: Give /AUTHORIZE the hash of the cluster of this node (Section 3.1), and check /GROUP.

BKSTATS, information unit *name*: sector *n*, *n* read-modify-write(s) needing *n* sector(s), *n* from the cache (in the last 2/16/128/512 written: *n*/*n*/*n*/*n*), *n* wait(s) for a range, *n* group fdatasync(s) for *n* write(s), *n* ZERO(s) (*n* as zeros written)

Explanation: Written on SIGUSR1 for each disk, after its UNITSTATS: the sector of the disk; how many writes were done as read-modify-write (a disk of 4096-octet sectors) and how many sectors they needed first; how many of those came from the cache of the 128 sectors last written; how many were among the last 2, 16, 128 and 512 written; how many writes waited for another one on the same sectors; with /SYNC=GROUP, how many flushes covered how many writes; how many ERASE commands came, and how many of them were done by writing zeros.

User Action: None.

BLKERR, warning *operation* from *name* (*systemid*), XCT_ID *id*: *reason*, dropped

Explanation: A chunk of data came from a node for no command of the server: the command is over, or the node sent the chunk twice. The server drops it. It does no harm to either side.

User Action: None if it is rare. If it is frequent, report it with a log of /TRACE.

BLKSTATS, information block transfers: *n* chunk(s) sent (*n* octets), *n* RET_CNF, *n* REQ_DAT, *n* chunk(s) taken (*n* octets), *n* dropped

Explanation: Written on SIGUSR1 after ACKSTATS: the chunks of data sent to the nodes (READ) and their octets, the confirmations the nodes sent for them, the requests for data the server sent (WRITE), the chunks taken in and their octets, the chunks dropped.

User Action: None.

CHCLOSE, warning channel to node *name* (*systemid*) is closed: *reason*

Explanation: The channel to a node is closed. The reason is one of:

```
BYE received, the node is going down     the node shuts down or crashes
restarted by the node                    the node started a new channel
no answer to the handshake in 5 seconds
nothing heard from the node for 8 seconds
the node has changed its name or system ID
the server is stopping
```

A node that is still up opens the channel again in a few seconds.

User Action: None if the channel opens again. If it does not, or if it closes again and again, see Section 7.3.

CHOPEN, information channel to node *name* (*systemid*) is open, CC V1.*n*, pipe quota *n*, buffer *n*, services *xxxx*/*xxxx*

Explanation: The handshake with the node is over. The message shows the version of the node, its pipe quota, its buffer size and its services (offered/required).

User Action: None.

CONCLOSE, information connection *name* of *node* (*systemid*) is closed: *reason*

Explanation: A connection to the disk server is closed: the node disconnected, its circuit closed, or the server ended it.

User Action: None; the node connects again. If the reason is “an end message was held back”, see MSCPERR.

CONOPEN, information connection *name* of *node* (*systemid*) to *server* is open, credit *n*/*n*

Explanation: An OpenVMS node has connected to the disk server MSCP$DISK: its disk class driver, VMS$DISK_CL_DRVR. The message shows the credits of both sides. The disks of the server now appear on the node.

User Action: None.

DEVBUSY, fatal the device *device* is in use: mounted on this host, or another server has it

Explanation: The server cannot open a block device of /UNIT*n* for itself: it is mounted on the Linux computer, it is in use by LVM or RAID, or another server has it. The server stops.

User Action: Dismount the device, or give another one. Never serve a device the Linux computer uses.

IDCLASH, fatal node *name* (*adapter*) uses the system ID low word *n* of this server

Explanation: Another node uses the same low 16 bits of the SCSSYSTEMID as the server. Two such nodes cannot be told apart on the LAN. The server stops and sends nothing.

User Action: Give the server another /SYSTEMID. On OpenVMS, `SHOW CLUSTER` or `MCR SYSGEN SHOW SCSSYSTEMID` shows the IDs in use.

IOERR, error *operation* of unit *name*, LBN *n*, *n* octets, failed: errno=*n* (*text*)

Explanation: The Linux file or device of a disk could not be read or written. The message shows the command, the disk, the block, the size and the error of Linux. The OpenVMS node gets a drive error.

User Action: Check the file system and the disk of the Linux computer. “No space left on device” means that the file is sparse and its file system is full.

IOSTUCK, warning *operation* of unit *name*, LBN *n*, is in the backend for *n* seconds: cancelled

Explanation: A command has been in the Linux disk for 15 seconds. The server cancels it; when the cancel takes, the node gets a drive error. When it does not - Linux is still writing - the disk waits for it.

User Action: Check the Linux disk under the unit: a dead disk, a lost iSCSI or NFS server. Look at `dmesg`.

IVPARAM, fatal illegal value *value* of /*name*: *what is expected*

Explanation: A qualifier has a wrong value. The message says what is expected. The server stops.

User Action: Correct the value.

LATSTATS, information unit *name*, *operation*: *n* command(s), average *n* us (backend *n*, transfers *n*, the rest *n*), longest *n* us; under 0.1/0.3/1/3/10 ms and longer: *n*/*n*/*n*/*n*/*n*

Explanation: Written on SIGUSR1 for each disk and for each of READ and WRITE, after its BKSTATS, when the disk has had such commands. The average time from a command to its answer, in microseconds, and how it was spent: the Linux disk (backend), the data on the LAN (transfers), and the rest - waiting for the disk, the server itself. Then the longest time, and how many commands took under 0.1, 0.3, 1, 3 and 10 ms and longer.

User Action: None.

LISTEN, information listening *n* second(s) before announcing the node

Explanation: The server listens for /LISTEN seconds before it sends anything.

User Action: None.

MANYNODES, warning more than *n* nodes on the LAN, node *name* is not remembered

Explanation: There are more nodes on the LAN than the server remembers (64). A node beyond that is not checked for a clash.

User Action: Report it.

MISALIGN, warning *device* starts *n* octets into a physical sector of *n*: every write is a read-modify-write of the disk

Explanation: A block device of /UNIT*n* - most often a partition - does not start on a physical sector of the Linux disk. No write of a node then fills whole sectors, and every one is a read-modify-write. The server still serves the device.

User Action: For a new disk, make the partition start on a multiple of 1 MB (`parted` and `fdisk` do so by default) and copy the data over.

MSCPERR, error end message of *operation* to *connection* of *node* (*systemid*) held back: *reason*

Explanation: The server was about to send an answer that would crash the OpenVMS node. It did not send it, and ended the connection. This is an error of the server.

User Action: Report it, with a log of /TRACE.

MSCPSTATS, information MSCP server: *n* command(s) in progress, *n* octets of data buffers in use

Explanation: Written on SIGUSR1 after BLKSTATS. When no node uses the disks, both numbers are 0.

User Action: None. A number that stays above 0 with no node connected is a fault of the server: report it with the log.

NAMECLASH, fatal node *name* (*adapter*, system ID *n*) uses the SCSNODE of this server

Explanation: Another node has the same SCSNODE as the server. The server stops and sends nothing.

User Action: Give the server another /NODE.

NEWNODE, information node *name*, system ID *n*, adapter *adapter*, CC V1.*n* (*n* octets)

Explanation: The server sees a node for the first time: its name, its system ID, its adapter and its version.

User Action: None.

NODESTATS, information node *name* (*systemid*): channel *state*, open *n* time(s); virtual circuit *state*, open *n* time(s); *n* connection(s) open

Explanation: Written on SIGUSR1 for each node the server knows, after UPTIME. OPEN twice and one connection is a node that uses the disks. A count of opens that grows means that the node comes and goes.

User Action: None. If the counts grow, see CHCLOSE and VCCLOSE in the log.

NODIRECT, information *path*: no direct I/O (*reason*), served through the page cache

Explanation: The file system of a file of /UNIT*n* does not allow direct I/O, or the size of the file is not whole blocks of 512 octets. The server serves the file through the page cache: it works, but every write costs more.

User Action: None; for speed, put the image on ext4 or XFS, or serve a partition or an LVM volume.

NOPARAM, fatal the parameter /*name* is mandatory

Explanation: A required qualifier is missing. The server stops.

User Action: Add it.

NOTOPIC, error sorry, no documentation on *topic*

Explanation: /HELP names a topic the help library does not have.

User Action: Give `mscphub /HELP` to see the topics.

SHUTDOWN, information signal *n* received, the server is stopping

Explanation: The server got SIGINT or SIGTERM. It sends HOSTSHUT on every open virtual circuit and, if it has announced itself, BYE to the group: the OpenVMS nodes close their circuits and channels to it at once.

User Action: None.

STARTUP, information MSCPHUB *ident*, node *name*, system ID *n*, group *n*, on *interface* (*address*)

Explanation: The server has started.

User Action: None.

STATS, information rx *n* frame(s) (*n* CC, *n* TR, *n* dropped, *n* own, *n* bad password), tx *n* frame(s), *n* error(s)

Explanation: Written on SIGUSR1 after the NODESTATS: frames received and sent, and send errors. Dropped frames are frames of other groups or for other nodes; that is normal.

User Action: None.

SYSERR, fatal *call* failed, errno=*n* (*text*)

Explanation: A Linux system call failed. The server stops. The most frequent ones:

```
socket(AF_PACKET), errno=1      the server is not run as root (CAP_NET_RAW)
ioctl(SIOCGIFINDEX), errno=19   there is no such interface
```

User Action: Correct what the call and errno name.

TRSTATS, information transport: *n* packet(s) sent, *n* retransmitted, *n* received, *n* duplicate(s), *n* out of order, *n* of an old channel; window up to *n*

Explanation: Written on SIGUSR1 after STATS: packets sent, retransmitted, received, duplicates, out of order, packets of an old channel, and the widest the send window has been.

User Action: None. A few retransmissions are normal; many "out of order" mean that frames from the nodes are lost (Section 7.5).

UNITCFG, information unit *name* is *path*, *n* block(s)*attributes*, SLUN *xxxx*

Explanation: At start-up, for every disk: its name on the OpenVMS nodes, its file or device, its size, its kind, sector and medium, and its MSCP unit number.

User Action: None.

UNITSTATS, information unit *name*: ONLINE to *n* host(s), *n* command(s) running, *n* waiting, *n* read(s) (*n* octets), *n* write(s) (*n* octets)

Explanation: Written on SIGUSR1 for each disk, after MSCPSTATS: to how many nodes the disk is online (mounted, or being mounted), how many commands run on it and how many wait for their turn, how many reads and writes it has done.

User Action: None.

UNSAFE, warning /SYNC=NONE: a WRITE is answered before its data is on disk - for tests only

Explanation: /UNSAFE or /SYNC=NONE is given.

User Action: Do not use it with data you need.

UPTIME, information MSCPHUB *ident*, node *name*, up *n* day(s) *hh*:*mm*:*ss*, OpenVMS time of *zone*

Explanation: The first message on SIGUSR1: the version, the name of the server, how long it has run and the zone of /TIMEZONE.

User Action: None.

VCCLOSE, warning virtual circuit to node *name* (*systemid*) is closed: *reason*

Explanation: The virtual circuit to a node is closed. The reason is one of:

```
the channel is closed ...                          see CHCLOSE
restarted by the node (START on the open path)     the node rebooted
HOSTSHUT received, the node is shutting down
no acknowledgement after 30 retransmissions        frames are lost
a block transfer the server does not take
the server is stopping
```

After a failure the server restarts the channel at once, and the node opens the circuit again in a few seconds.

User Action: None if the circuit opens again. If it closes again and again, see Section 7.5.

VCOPEN, information virtual circuit to node *name* (*systemid*) is open: *software version* on *hardware*, datagram *n*, message *n*

Explanation: The node has answered the server, and `SHOW CLUSTER` on it now shows the server. The message shows what the node told of itself: its software, version and hardware, its largest datagram and message.

User Action: None.

Chapter 7. Troubleshooting

Read the last messages of the log first:

```
# journalctl -u mscphub -n 50
```

`mscphub "/HELP=Messages`*NAME*`"` says what a message means and what to do; so does Chapter 6.

7.1 The Server Stops at Once

The last message is of severity F and says why:

Message	What to do
`F-NOPARAM`, `F-IVPARAM`	Correct the qualifier it names in the parameter file
`F-SYSERR, socket(AF_PACKET), errno=1`	Run the server as root
`F-SYSERR, ioctl(SIOCGIFINDEX), errno=19`	Correct the name of /INTERFACE
`F-IDCLASH`, `F-NAMECLASH`	Give another /SYSTEMID or /NODE
`F-DEVBUSY`	The device is mounted or used on Linux

A qualifier misspelled in the file gives `Skip unrecognized option`: it is not used, and the server runs without it.

7.2 No Node Is Seen

`ANNOUNCE` says 0 other nodes, and no `NEWNODE` comes.

1. Check /GROUP.
2. Check that the interface is up and in the right bridge or VLAN.
3. On OpenVMS, check that PEDRIVER runs: `MCR SCACP SHOW CHANNEL`.
4. Look at the wire:

```
# tcpdump -i eth1 ether proto 0x6007
```

7.3 No Channel Opens, or It Opens and Closes Again

The nodes are seen, but no `CHOPEN` comes:

- Look for `AUTHFAIL`: the hash of /AUTHORIZE is not that of the group.
- Check that /PASSIVE is not given.
- On OpenVMS, `MCR SCACP SHOW CHANNEL` shows the channel to the server and its state.

`CHOPEN` and `CHCLOSE` again and again: read the reason in `CHCLOSE`. “nothing heard from the node” means that frames are lost between the node and the server: check the switch, the bridge or the VLAN.

7.4 The Disks Do Not Appear on the Node

The channel and the circuit are open, but there is no `CONOPEN`.

- Check that the parameter file has /UNIT*n* lines (`UNITCFG` at start-up).
- On OpenVMS, check that the process CONFIGURE runs (`SHOW SYSTEM`): it loads the disk class driver.

If the channel is open but no `VCOPEN` comes, run the server with /TRACE for a short time and look for `PPD START` in the log, and for a warning after it.

7.5 The Circuit Closes, or Writes Are Slow

Read the reason in `VCCLOSE`. Send `SIGUSR1` and look at `TRSTATS`:

- many retransmissions mean that frames to the nodes are lost;
- many “out of order” mean that frames from the nodes are lost.

Writes slow and reads fast is the second case: a node sends a lost frame again only after a second or more. Check the switch, the bridge, the VLAN, the cables.

7.6 A Command Fails on the Node

`MOUNT` or `COPY` fails with an I/O error. Look for:

Message	What it means
`E-IOERR`	The Linux file or disk failed: full, read only, broken
`W-IOSTUCK`	The Linux disk hangs: look at `dmesg`
`E-MSCPERR`	An error of the server: report it
`W-BLKERR`	A stray chunk of data; harmless if rare

7.7 The Disks Are Slow

Message or sign	What to do
`I-NODIRECT`	Move the image to ext4 or XFS
`W-MISALIGN`	Make a new partition on 1 MB
A sparse image	Write it full once (Section 3.2.1)
Many read-modify-writes in `BKSTATS`	INITIALIZE the 4K disk with /CLUSTER=16 (Section 4.3)

`LATSTATS` shows where the time of a command goes: in the Linux disk (backend) or on the LAN (transfers).

7.8 Collecting Information for a Report

```
# systemctl kill -s USR1 mscphub
# journalctl -u mscphub --since '1 hour ago' > mscphub.txt
```

Add the parameter file, without the line /AUTHORIZE. For a protocol problem, run the server with /TRACE for the minute in which the problem happens, and add that log too.

Appendix A. Files

File	Purpose
`/opt/mscphub/sbin/mscphub`	The server
`/etc/opt/mscphub/mscphub.conf`	The parameter file of the service; yours, never overwritten
`/etc/opt/mscphub/mscphub.conf.example`	A sample parameter file
`/lib/systemd/system/mscphub.service`	The systemd unit
`/opt/mscphub/share/man/man8/mscphub.8`	The manual page
`/opt/mscphub/share/help/mscphub.hlb`	The help library, English
`/opt/mscphub/share/help/ru/mscphub_ru.hlb`	The help library, Russian
`/opt/mscphub/share/doc/mscphub/`	The manuals

Environment Variables

Variable	Meaning
`MSCPHUB_HELPLIB`	The help library /HELP reads, in place of the installed one

Appendix B. Limits

Item	Limit
Disks per server	8 (`/UNIT0` to `/UNIT7`)
Name of a disk	`DK`*c**n*: a letter A to Z, a number 0 to 9999
Size of a disk	The size of the file or the device, in blocks of 512 octets
Sector of a Linux device	512, 1024, 2048 or 4096 octets
Nodes on the LAN remembered	64
Commands in progress per node	/CREDITS, 2 to 64; 16 by default
Data buffers of all commands	16 MB; a command that finds none waits
Packets unacknowledged per circuit	16
Time a disk command may take	15 seconds, then IOSTUCK
Retransmissions before a circuit fails	30
LAN adapters	1